

Let's begin with the most versatile data structure in Python: the list.

Lists: Your Trading Timeline

What Are Lists?

A list in Python is an ordered collection of items, similar to a shopping list or a playlist. Lists are perfect for storing sequences of data that have a natural order, like:

- Historical stock prices ordered by date
- Trading signals across multiple timeframes
- Portfolio positions ranked by size
- Technical indicators calculated over time

Creating Lists: Building Your Data Collection

Creating a list in Python is straightforward—simply place comma-separated items inside square brackets:

```
1. # A list of stock prices for Apple over 5 days
2. apple_prices = [142.56, 143.78, 144.22, 143.86, 145.03]
3.
4. # A list of stock symbols in a portfolio
5. portfolio = ["AAPL", "MSFT", "GOOGL", "AMZN", "TSLA"]
6.
7. # A list of different data types (mixed content)
8. trading_data = ["AAPL", 145.03, 1000, True]
9.
```

As you can see, lists can contain numbers, strings (text), or even a mix of different types. This flexibility makes them incredibly versatile for organizing trading data.

Accessing List Elements: Finding Specific Data Points

Once you have data in a list, you'll need to access specific items. Python uses indexing to access list elements, with the first item at index 0:

```
1. prices = [142.56, 143.78, 144.22, 143.86, 145.03]
```